





Noise Model (I)

Background: Noise and relevance for NISQ era

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What are we covering in this module?

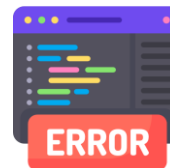
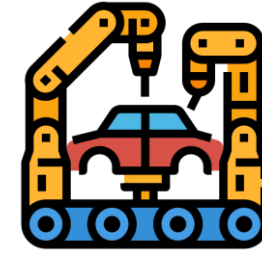
- Noise models
- Quantum error mitigation
- Quantum volume and performance matrices
- Hybrid quantum classical computing

What are we covering in this chapter?

- **Noise models: Noise and relevance for NISQ era**
 - *Background on noise: What is noise in systems*
 - *Different noise sources in quantum hardware*
 - *Why is noise modeling important?*
 - *What are NISQ devices?*
- Noise models: Modeling noise in Quantum

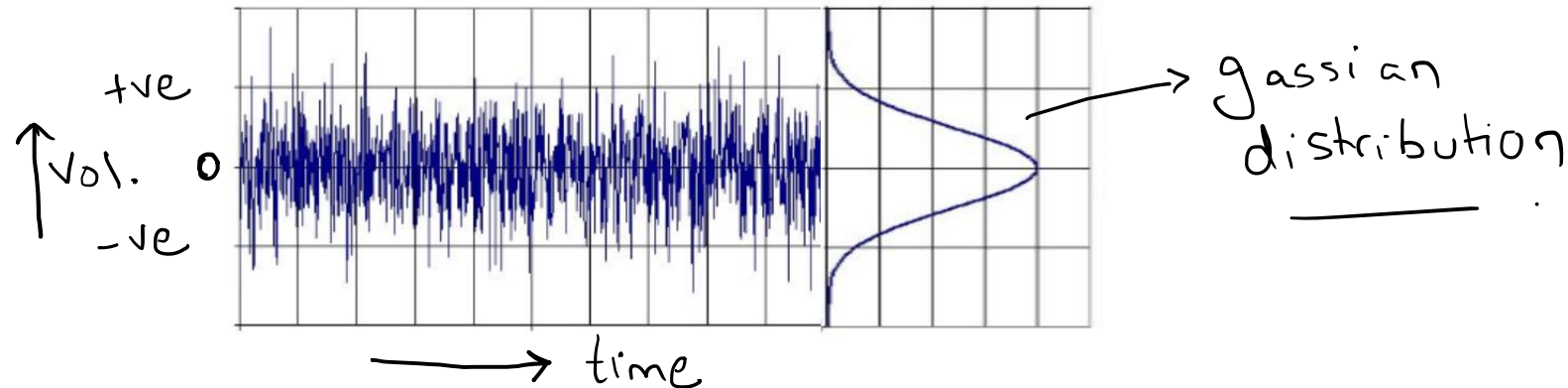
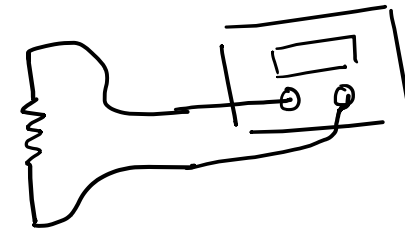
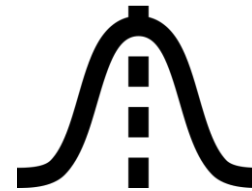
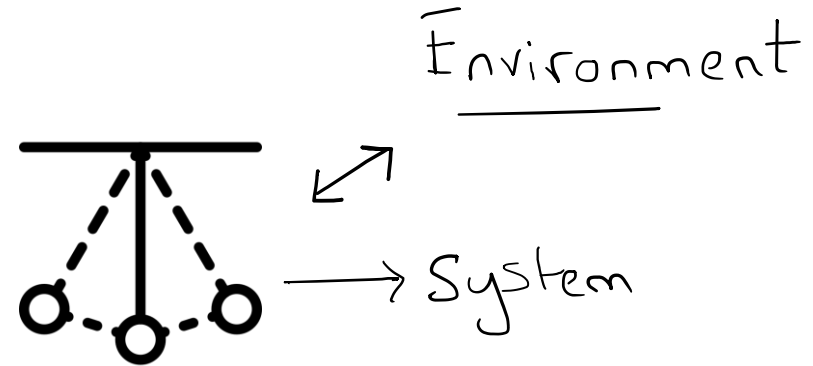
Background on Noise

- Types of errors in a system:
 - Systematic errors – Deterministic: Ex. gain error, offset error
 - Random errors – Stochastic: Ex. Thermal noise
 - Blunders errors – Outright mistakes

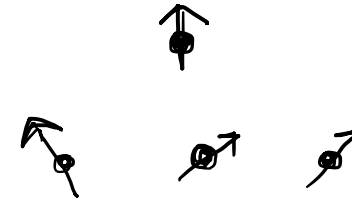


Background on Noise

- Characterizing classical noise:
 - Modeling the environment
 - Modeling the interaction with the system
- Description:
 - Mean value
 - Standard Deviation
- Example: Thermal noise (0, σ)

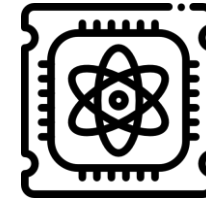


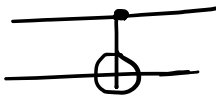
- Quantum noise: Uncertainty in its physical quantity
- Closed and Open systems
- Loss of quantum information: decoherence
- Example: Spin qubit interacting with other spins



Noise at quantum hardware

- Device noise
 - Superconducting qubits: $1/f$ flux noise in SQUIDs
 - Spin qubits: Interaction with nuclear spins



- Gate noise 

- Readout/measurement noise 

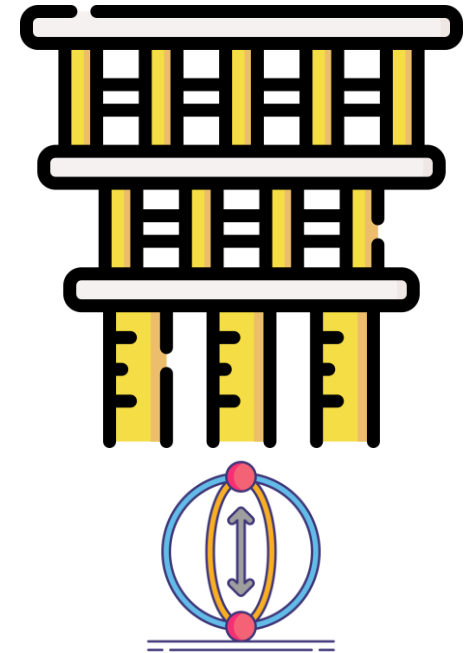


Why is noise modeling important?

- As a Quantum Hardware designer:
 - To understand the limitations of your hardware and model/compensate accordingly
 - To give specifications of your hardware to customers
 - To find ways to mitigate error
- As an end-user for quantum computers:
 - To make sense of your operations
 - To design efficient algorithms that can overcome the limitations

What are NISQ devices?

- NISQ: Noisy Intermediate Scale Quantum
- Few qubit systems (~ 50 -100)
- Noisy qubits
- Hybrid quantum-classical
 - VQE
- Noise becomes even more important!





Thank you for your attention!

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- J. Preskill, "Quantum Computing in the NISQ era and beyond", Quantum, vol. 2, p. 79, 2018. Available: [10.22331/q-2018-08-06-79](https://arxiv.org/abs/1808.07331).
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